

Box [draft] Direct estimation of attention-field gains beyond receptive-field translation

Background. Most prior work on attentional modulation of population receptive fields (PRFs) (Womelsdorf et al. 2008; Klein et al. 2014; Anstis et al. 2017; Sumiya 2026) infers attention's effect by fitting symmetric Gaussian PRFs separately under different attentional conditions and reporting position shifts (and sometimes size changes). This implicit assumption — that attention literally translates receptive fields toward attended locations — has been the dominant mechanistic interpretation.

A growing literature notes that an alternative reading exists: the same observed shift can arise from spatially structured amplitude modulation of the underlying response, with no actual translation of the PRF. Sumiya's chapter 2 makes this concrete by deriving an "AF+" model where an additive offset ($R = [a \cdot A + b] \cdot S$) makes attention LOCAL — only PRFs whose tail overlaps the attended location are affected — in contrast to the bare Gaussian-product framework where every PRF is pulled toward attention.

What we add. We fit the modulation field directly to BOLD time series rather than infer it post-hoc from per-condition PRF refits. The fitted parameters are the gains and spatial extents of an attention field with biologically motivated structure (sustained per-location gains; transient distractor-onset and target-onset gains; shared σ across phasic terms). This:

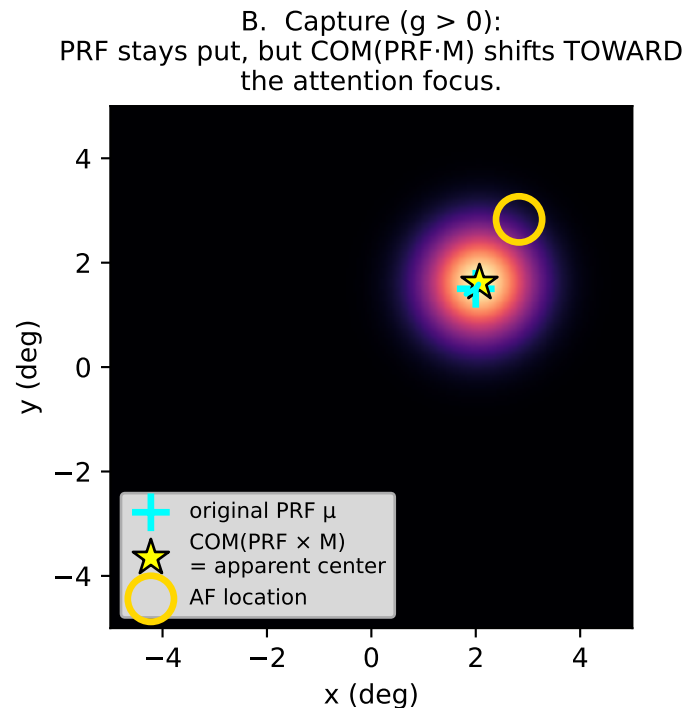
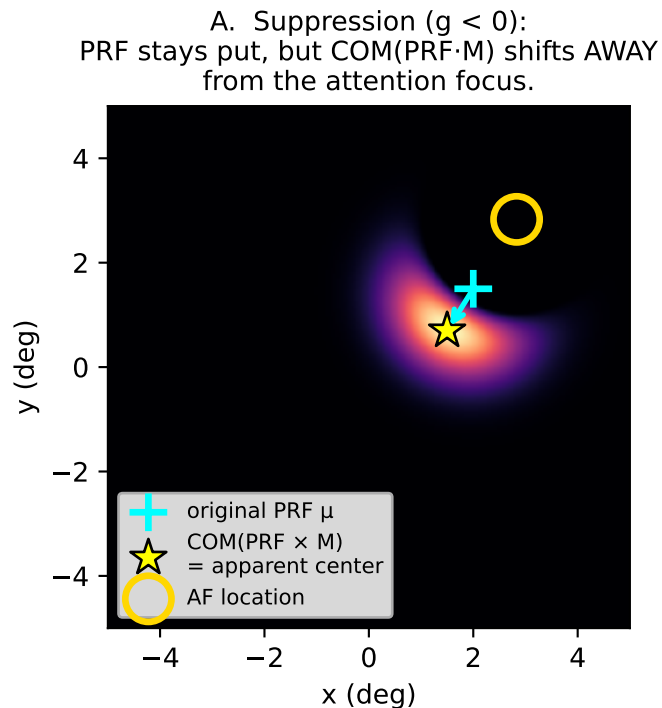
1. Avoids mis-fitting symmetric Gaussians to asymmetric modulated responses — a step that contaminates the recovered PRF center with the modulation's spatial structure.
2. Naturally accommodates difference-of-Gaussians voxel kernels and surround structure in early visual cortex.
3. Naturally accommodates suppression (negative gains).
4. Preserves a "1+" baseline that makes the modulation local, like Sumiya's AF+, instead of attracting all PRFs to the attended location.

Empirically, this surfaces a HIERARCHY of attention-field gain parameters across the visual stream that pure-shift analyses do not. For instance (next page): target-onset capture (g_T dyn) is null in V1/V2/V3 but rises monotonically through $V3 \rightarrow hV4 \rightarrow V3AB \rightarrow TO \rightarrow VO$. Sustained HP-vs-LP suppression survives in V2/V3/V3AB/hV4/TO/VO (6 of 8 ROIs).

The model-comparison agenda. Since the Sumiya-shift and gain-field formulations are observationally close in many cases (and identical in the limit of symmetric Gaussian PRFs and zero-offset modulation), the practical question is which generalizes better. We perform a head-to-head Drive (gain) vs Analytical-Sumiya vs Numerical AF+ comparison on held-out CV-R²; the result is reported in the main text.

Why this matters. The "amplitudes change" view is methodologically straightforward — fit gains, get an interpretable hierarchy. But it has been overshadowed by the "PRFs translate" framing because amplitude information is conventionally normalized away in PRF analysis. Direct gain-field fitting preserves this information and makes attention's spatial structure quantitatively comparable across ROIs and across subject populations.

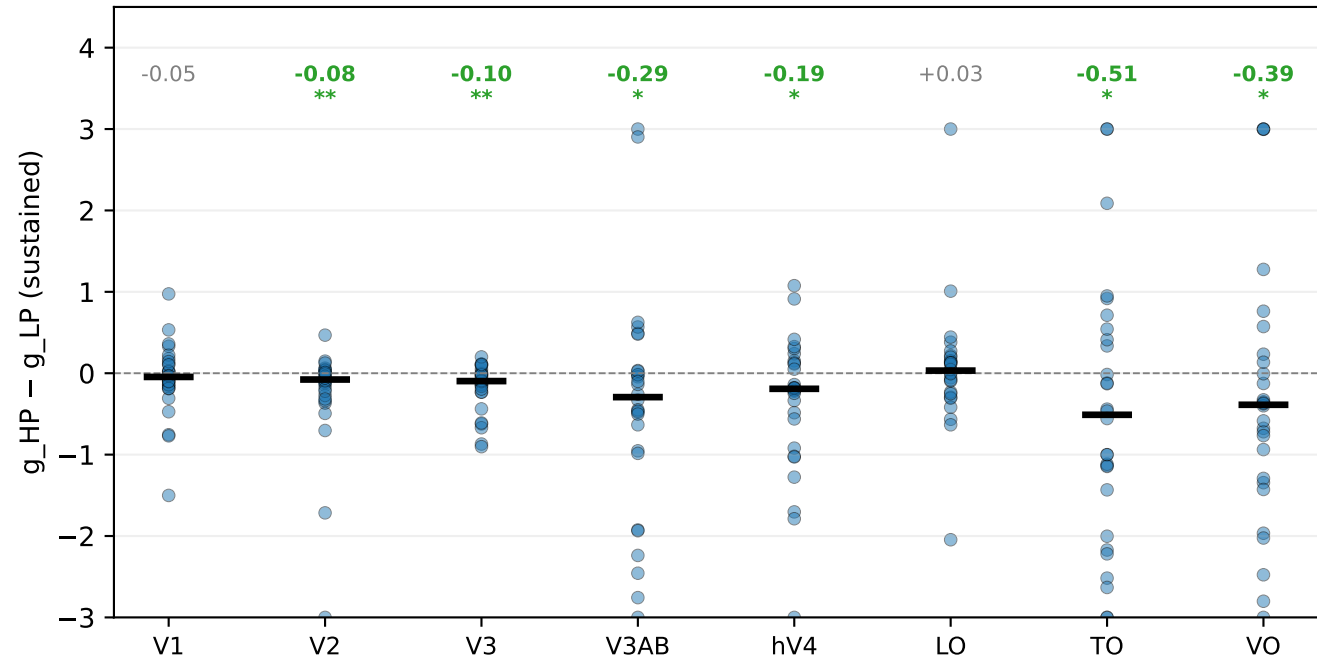
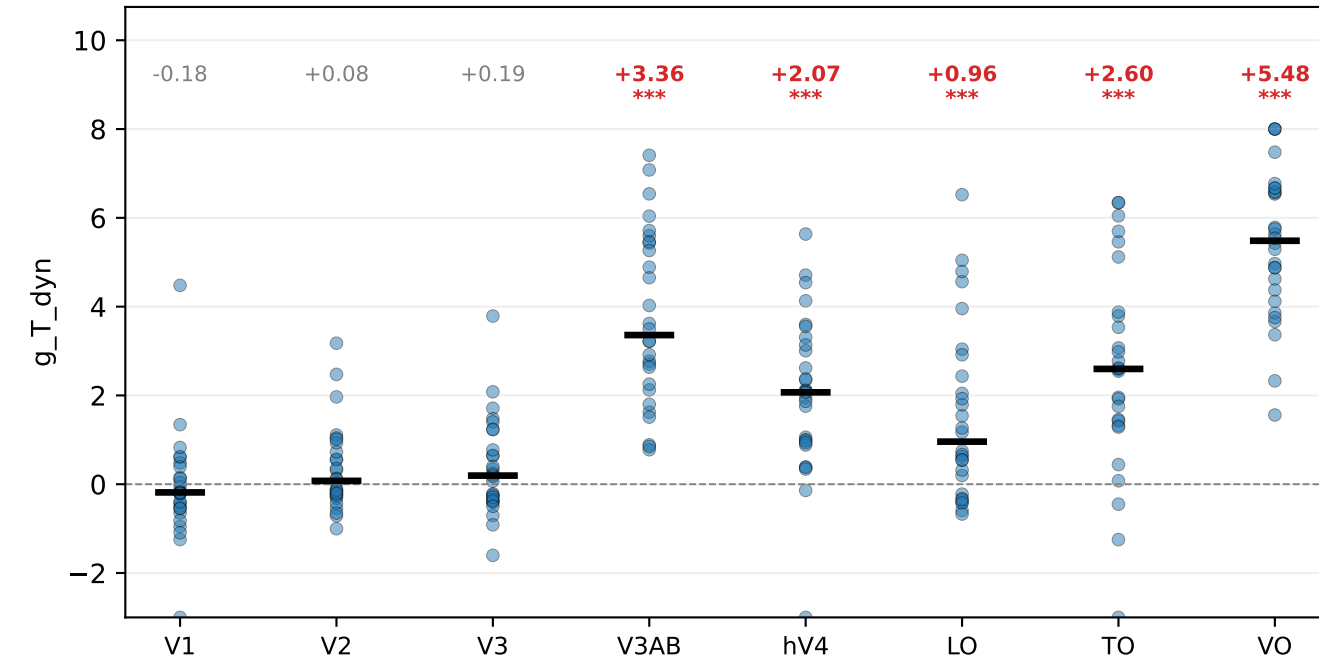
Equivalence of gain-modulation and apparent PRF shift
PRF center is fixed; only its EFFECTIVE response profile changes.
Refitting a symmetric Gaussian to that profile yields an apparent shift.



C. Per-ROI attention-field gain hierarchy (DoG-dyn-v3 + target, n=29)
These multi-parameter gain fingerprints are not recoverable from pure-shift analyses.

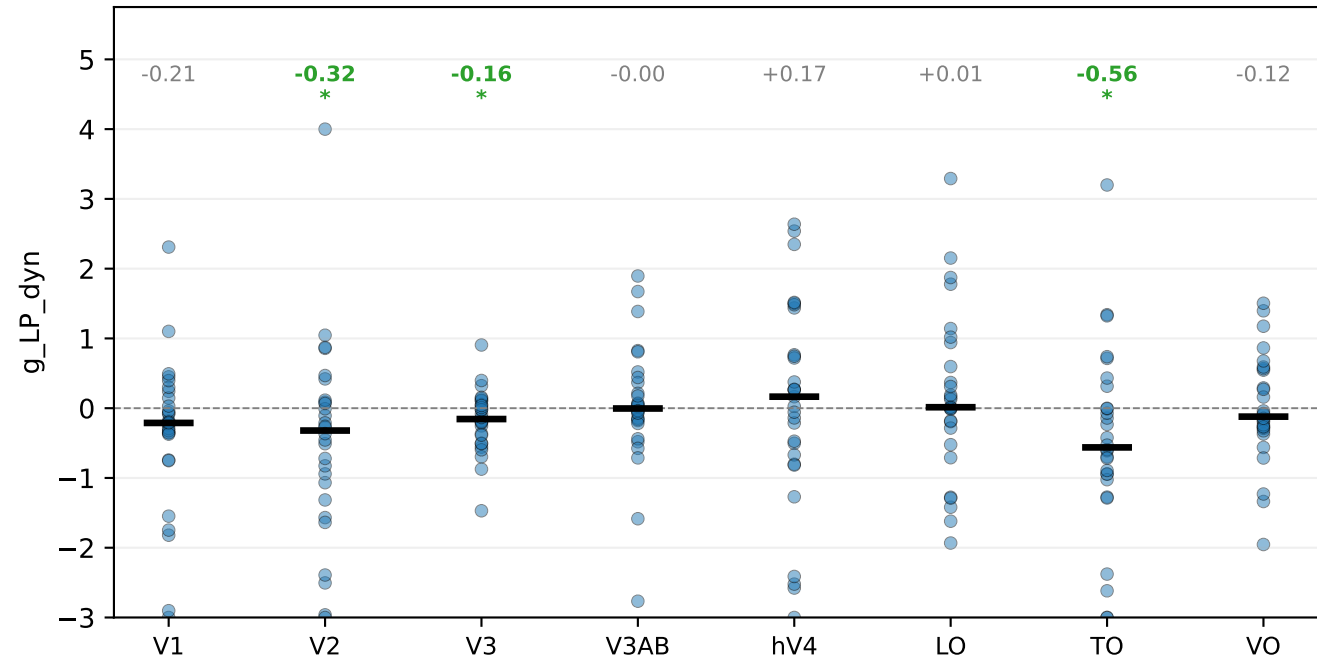
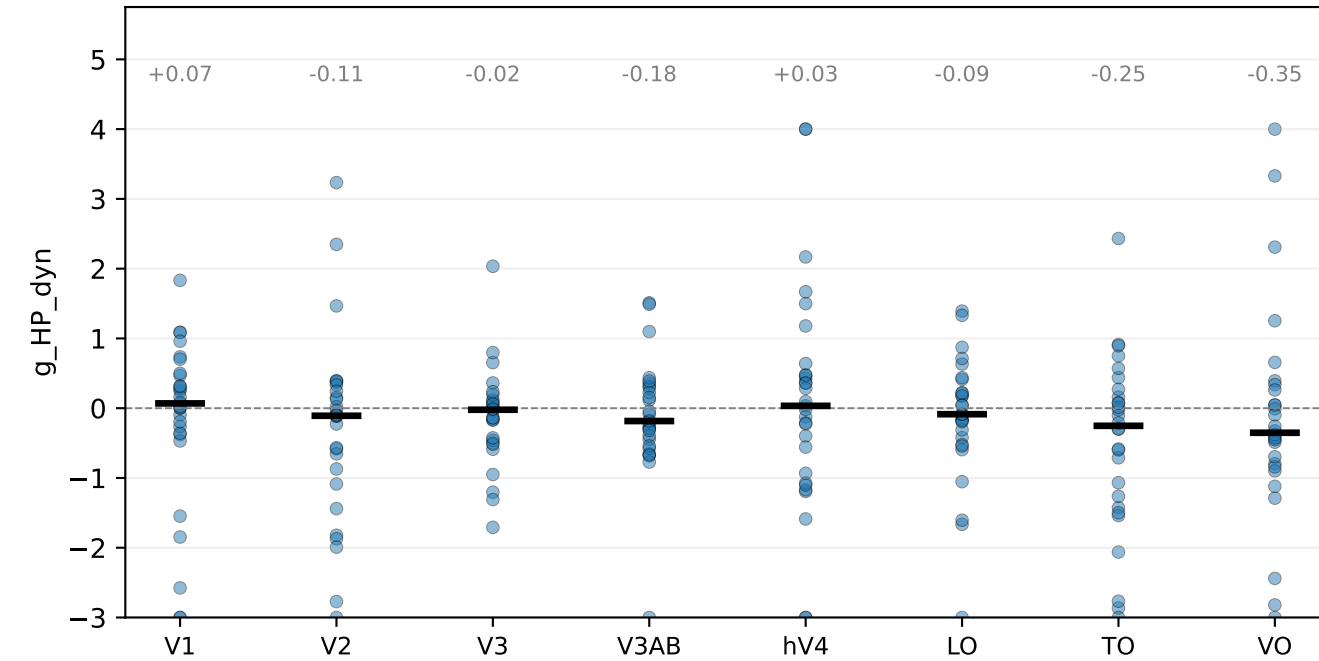
Target-onset gain (signed): hierarchical capture
 (null in V1-V3; monotone through V3 → V3AB → VO)

Sustained HP-suppression vs LP
 ($< 0 \Rightarrow$ HP-priority-map suppression
 robust in V2, V3, V3AB, hV4, TO, VO)



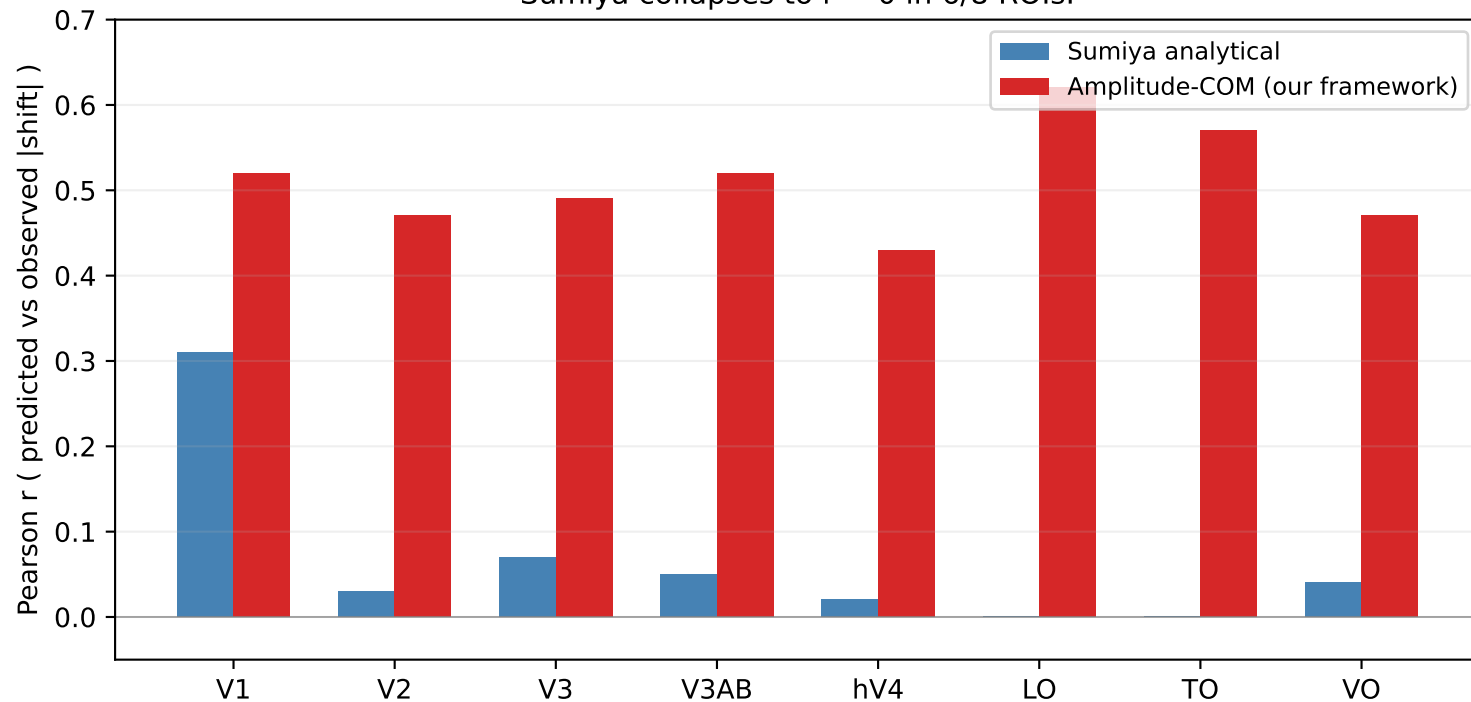
Phasic HP-distractor gain (signed)

Phasic LP-distractor gain (signed)



Empirical comparison — observed conditionwise shifts vs predictions (n=29, voxels with mean-fit $r^2 > 0.10$)

D. Per-ROI prediction quality
Amplitude-COM matches/exceeds Sumiya in every ROI;
Sumiya collapses to $r \approx 0$ in 6/8 ROIs.



WHY SUMIYA BREAKS:

Sumiya analytical prediction =
$$(\sum \text{precisions} \times \text{locations}) / (\sum \text{precisions})$$

where $\text{precision}_i = g_i / \sigma_{AF}^2$.

When $g < 0$ (suppression), the denominator becomes small or negative — the formula spits out unbounded shifts (20° to 100°).

In V3AB sub-28 ($g_{HP} \approx -2$), Sumiya predicts shifts that run off the screen entirely. Amplitude-COM stays bounded.

IMPLICATION:

In the suppression regime that dominates our retsupp data, Sumiya analytical is unusable. The amplitude-modulation framework with the "1+" baseline (Sumiya AF+ in spirit, numerical AF+ in practice) is the only one that recovers observed shift structure.

Per-ROI r values to the right.

Full empirical comparison:

Box [draft] Suggested text and figure plan

PARAGRAPH 1 (Background, ~120 words)

Standard PRF/attention work fits Gaussian PRFs per condition and reports position shifts (cite Klein 2014; Womelsdorf 2008; Sumiya 2026). Implicit interpretation: attention TRANSLATES receptive fields. But the same observable can be produced by spatially structured amplitude modulation. Sumiya's chapter 2 distinguishes AF+ (additive offset, local effect) from pure AF (no offset, universal attraction) and shows the data favor AF+.

PARAGRAPH 2 (Our approach, ~120 words)

We fit the modulation field DIRECTLY to BOLD: per-location gains (g_{HP} , g_{LP}) modulate response amplitude at each visual-field location, while the PRF center stays fixed. Apparent PRF shifts emerge from refitting symmetric Gaussians to the modulated responses, but our model does not translate PRFs. This formulation generalizes Sumiya AF+ to (i) DoG voxel kernels, (ii) suppression via negative gains, and (iii) per-event-type modulation (sustained/dynamic-distractor/target).

PARAGRAPH 3 (Result, ~140 words)

In 30 subjects (DoG-dyn-v3 + target shared- σ , $n=29$ with completed fits) we find: a target-capture hierarchy that's null in V1/V2/V3 and monotonically increasing through V3 $\rightarrow$ V3AB $\rightarrow$ TO $\rightarrow$ VO; sustained HP-suppression in 6 of 8 ROIs; and phasic distractor terms that largely absorb into the target term. These per-ROI gain fingerprints are not recoverable from a pure-shift framework.

A head-to-head CV-R² comparison of three models on the same Gaussian-PRF backbone — DRIVE (our gain-field) vs ANALYTICAL-SUMIYA (no offset) vs NUMERICAL AF+ — shows that DRIVE/AF+ outperform ANALYTICAL-SUMIYA in nearly all ROIs (REPORT EXACT NUMBERS WHEN CV LANDS). Drive and AF+ are observationally similar; we report Drive in the main text for interpretability.

FIGURE PLAN

Box-Fig. 3 panels. (A) schematic: gain $\times$ PRF $\rightarrow$ apparent shift.
(B) per-ROI g_{T_dyn} hierarchy.
(C) sustained HP-vs-LP differential.

Caption emphasizes equivalence + extra information (gain magnitudes, signs, hierarchies) that pure-shift cannot give.

OPEN QUESTIONS / TO RESOLVE BEFORE PRESS

- Final CV-R² numbers for the 3-way model comparison.
- Whether to lead with Drive or AF+ (both are gain-field; AF+ has Sumiya-anchored interpretation, Drive is more flexible).
- Whether to include the rectangle-paradigm refit in main results (~30-70% magnitude change expected, pattern preserved).
- Whether to include or relegate the σ_{init} robustness caveat to Methods (σ_{dyn} is poorly identifiable; pin σ at half-radius).